



Stroke and Disability: Incidence, Risk Factors, Management, and Impact

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ABSTRACT

Stroke is a leading cause of disability worldwide, leaving a significant portion of the population grappling with its long-term effects. The incidence of stroke is 3.6 on average higher compared to high-income countries. This study explores the global and Saudi Arabian burden of stroke-induced disability, highlighting its prevalence, risk factors, and the need for comprehensive rehabilitation strategies. The primary source of information was the international trusted databases, including Web of Science, Scopus, PubMed, and the websites and reports from international organizations such as the International Atomic Energy Agency, World Stroke Organization, and World Health Organization. The study revealed that women, especially after menopause, should be aware of their increased stroke risk and take steps to manage their risk factors such as high blood pressure, diabetes, and high cholesterol. Managing established risk factors such as high blood pressure, diabetes, and cholesterol remains the cornerstone of stroke prevention for all individuals, regardless of profession. Workplace wellness programs that promote healthy lifestyles and stress management can be beneficial for employees in any occupation. Overall, stroke presents a growing public health challenge in Saudi Arabia. While the incidence may be lower than in some high-income countries, it is on the rise. More research is needed to gain a clearer picture of stroke-related disability, develop effective prevention and treatment strategies, and solidify the connection between specific occupations and stroke incidence.

KEYWORDS

stroke, disability, risk factors, rehabilitation

INTRODUCTION

Stroke, cerebrovascular accident, or brain attack is one of the leading causes of disability worldwide, and the incidence is increasing worldwide, with ischemic stroke representing 80% of stroke incidence (GBD 2019 Stroke Collaborators, 2021). The annual stroke incidence has increased noticeably in the last two decades and is expected to increase due to an increase in obesity. Currently, 25% of adults over 25 will have a stroke in their lifetime (GBD 2019 Stroke Collaborators, 2021; Rahman et al., 2021). The improvement of stroke prevention measures contributed to the reduction of stroke-related disability in developed countries from 2010 to 2019 (Foerch et al., 2022). Stroke is a cerebrovascular condition that occurs when blood flow to a part of the brain is interrupted, leading to the death of brain cells due to oxygen

deprivation and subsequent neurological deficits (Rahman et al., 2021). Permanent brain cell damage will happen in a few minutes to hours in cases of stroke due to sudden bleeding. Lack of oxygenation can lead to cell death and damage, resulting in physical, cognitive, and emotional impairments (GBD 2019 Stroke Collaborators, 2021; Feigin et al., 2022). Quick treatment is crucial to save brain cells and reduce damage. “Time is brain” signifies the importance of immediate treatment upon detecting signs of stroke and notifying emergency services (GBD 2019 Stroke Collaborators, 2021; Rahman et al., 2021; Feigin et al., 2022). Stroke survivors frequently have severe mental and physical disabilities that place a significant social and economic burden on them. According to the World Health Organization (WHO),

stroke is the second leading cause of death globally and a major contributor to disability (WHO, 2022). It is estimated that over 101 million people worldwide live with stroke-related disability, with millions more experiencing recurrent strokes (WHO, 2022; WSO, 2022). Strokes can be classified into two main types: ischemic and hemorrhagic. Ischemic strokes occur when blood flow to the brain is disrupted due to a blocked vessel, while hemorrhagic strokes are caused by bleeding in the brain or the space around it. Hemorrhagic stroke is less common but carries a higher death rate, representing 10-20% of strokes yearly (An et al., 2017). The prevalence of hemorrhagic stroke is more common in males and rises with the time of life. The risk factors for hemorrhagic stroke include high blood pressure, aneurysms, arteriovenous malformations (AVMs), blood-thinning medications, and head trauma (GBD 2019 Stroke Collaborators, 2021; Rahman et al., 2021; Feigin et al., 2022). Intracerebral hemorrhage (ICH) is the second most common subtype of stroke and a critical disease, usually leading to severe disability or death with an incidence of 29.90 per 100,000 individuals per annum (An et al., 2017). ICH is more common in Asians, advanced age, male gender, and low- and middle-income countries. The death rate from stroke is a function of the availability of prompt medical intervention. It was reported that annually, over 15 million people worldwide suffer a stroke (GBD 2019 Stroke Collaborators, 2021; Feigin et al., 2022; WHO, 2022; WSO, 2022) with 5 million mortality rates, and another 5 million are left permanently disabled (Feigin et al., 2022). The higher death due to stroke in lower and lower-middle-income countries experiences a disproportionate share of the stroke burden, with 86% of stroke deaths and 89% of disability (Feigin et al., 2022). The stroke risk factors include age, high blood pressure, diabetes, atrial fibrillation, carotid artery disease, smoking, obesity, and physical inactivity (Rahman et al., 2021; Feigin et al., 2022). Also, stroke can be caused by hemorrhage, in which a stroke is caused by bleeding in the brain due to a ruptured blood vessel; this will result in pressure and damage to brain tissue. Risk factors for stroke include high blood pressure, high cholesterol, diabetes, and atrial fibrillation. Besides, lifestyle factors such as smoking, physical inactivity, and obesity increase the likelihood of stroke. Some genetic factors, such as a family history of stroke, may increase the risk of experiencing a stroke. Several factors increase the risk of stroke and its subsequent disability (Rahman et al., 2021; Feigin et al., 2022). These include traditional cardiovascular risk factors such as high blood pressure, diabetes, high cholesterol, and smoking. Additionally, unhealthy lifestyle choices such as physical inactivity, unhealthy diet, and excessive alcohol consumption contribute to stroke risk. Age is another significant factor, with the risk of stroke increasing considerably after the fifth decade of age. Given that stroke is one of the primary causes of disability globally, a sizable percentage of stroke survivors in Saudi Arabia are probably disabled to some extent. Stroke may impact speech, mobility, cognitive abilities, and general independence (GBD 2019 Stroke Collaborators, 2021; Rahman et al., 2021; Feigin et al., 2022). The Middle East, which includes Saudi Arabia, has an especially alarming stroke rate. The nation has a high rate of obesity and diabetes, two risk factors that contribute

to an increased incidence of stroke. According to studies, stroke is one of the main causes of mortality and disability in Saudi Arabia, and a sizable fraction of stroke survivors live with long-term disabilities. According to estimates, there are 30,000 new cases annually (Robert and Zamzami, 2014; Alqahtani et al., 2020; Gardener et al., 2020; Basri et al., 2021). According to one study, Saudi Arabia has a 29-57 case incidence of stroke per 100,000 person-years (Robert and Zamzami, 2014; Alqahtani et al., 2020; Gardener et al., 2020). In general, stroke incidence is lower in low-income nations than in high-income ones. However, because of changes in lifestyle and the rising prevalence of risk factors, including diabetes and hypertension in the Kingdom of Saudi Arabia (KSA), the incidence is rising (Bakraa et al., 2021). In addition to the disability and its consequences, stroke has a detrimental impact on people's quality of life and imposes a heavy financial burden (Donkor, 2018). Every year, there are more incidences of stroke, which presents a significant burden to the healthcare system in the KSA and globally. Although there is information on the prevalence of strokes, there are little data on stroke-related disability in the KSA. The article enumerates the present state of knowledge on the global elements that influence stroke incidence, diagnosis, and management, with special emphasis on Saudi Arabia.

MATERIALS AND METHODS

The relevant literature regarding stroke incidence, causes, impact, and management was carefully collected and summarized in the international literature with a specific emphasis on Saudi Arabia. The primary source of information was the international trusted databases, including Web of Science, Scopus, PubMed, and the websites and reports from international organizations such as the World Stroke Organization (WSO), WHO and national databases including Saudi Health Reports, and relevant reports from the United States. The study covered the published studies from 2011 to 2024. The manuscript covered all published papers and reports in the last 10 years and was published in English. Scientific articles not included in the databases mentioned above were excluded.

RESULTS AND DISCUSSION

Stroke incidence

Stroke incidence and ethnic group

Stroke incidence varies among different ethnic groups. Previous studies suggested that Black and South Asian populations are generally considered to be at greater risk for stroke up to two times compared to the White population (Robert and Zamzami, 2014; Gardener et al., 2020). In addition to that, Black and South Asian populations have a stroke incidence at a younger age compared to the White and other populations (Fluck et al., 2023; Kalasapudi et al., 2023). Many factors contribute to the disparity:

- I. Socioeconomic factors: Lower socioeconomic status can limit access to quality healthcare, healthy food options,

and safe living environments. These factors contribute to risk factors such as high blood pressure, diabetes, and obesity, which are more prevalent in certain ethnic groups.

- II. Genetic predisposition: Certain ethnic groups may have a genetic predisposition to conditions that increase stroke risk, such as sickle cell disease or hypertension.
- III. Cultural and lifestyle factors: Dietary habits, physical activity levels, and access to preventive healthcare can vary among ethnic groups, and these factors can influence stroke risk.

Therefore, it is essential to note that data on stroke incidence by ethnicity can vary depending on the population studied and the methodology used. Ethnicity is just one factor influencing stroke risk. Age, family history, and lifestyle choices also play a significant role.

Stroke incidence and gender

Stroke affects both men and women, but incidence rates have some interesting differences. Previous studies showed that females are more likely to experience a stroke than males (Roger et al., 2011; Ospel et al., 2022). However, men tend to have strokes earlier in life compared to women. The average age of the first stroke is 68.6 years for men and 72.9 years for women (Ospel et al., 2022). The possible causes are due to hormonal fluctuations: Estrogen may have some protective effects against stroke in pre-menopausal females. After menopause, this protection is lost, potentially contributing to the higher incidence in women. Pregnancy complications are also considered a potential factor because certain pregnancy complications, like preeclampsia, can increase a woman's risk of stroke later in life. In addition to that, socioeconomic factors are likely to affect stroke incidence because females may have less access to preventive healthcare or be more likely to prioritize the health of family members over their own, potentially increasing their stroke risk (Roger et al., 2011; Ospel et al., 2022). Stroke severity: Studies suggest that strokes in women may be more severe and have a higher mortality rate compared to strokes in men. Stroke subtypes: Men are more likely to experience hemorrhagic strokes (bleeding in the brain), while ischemic strokes (blood clots blocking blood flow) are more common in women. When considering age-specific rates, the picture becomes more complex. In younger age groups (<30 years), women have a slightly higher stroke incidence than men (Ospel et al., 2022). During midlife (30-50 years), men have a higher stroke incidence. In later life (>50 years), the incidence rates become more similar, with women surpassing men (Ospel et al., 2022).

Stroke and occupation

The relationship between stroke incidence and the type of occupation is complex. Compared to other well-established risk factors such as high blood pressure and diabetes, there is limited conclusive evidence on the direct causal link between specific occupations and stroke incidence. However, some

occupations or work habits have higher risks because they involve physical exertion, which includes jobs demanding heavy physical labor or lifting, which may increase cardiovascular stress and potentially contribute to stroke risk. Jobs with long working hours and shift work can disrupt sleep patterns, raise stress levels, and promote unhealthy lifestyle habits, all of which can contribute to stroke risk factors such as high blood pressure and obesity. Finally, exposure to certain chemicals or toxins may be a risk factor, although more research is needed to confirm this association (Ovbiagele and Nguyen-Huynh, 2011; Arnao et al., 2016). Therefore, the increased risk associated with particular occupations may be relatively small. Workplace factors such as stress levels, safety regulations, and access to health screenings can influence stroke risk within an occupation.

Stroke incidence and family history

Age, family history, and overall health status likely play a more significant role in stroke risk. Having a close family member (parent, sibling, child) with a history of stroke increases the risk probability of an individual experiencing a stroke. The strength of the association between family history and stroke risk varies depending on several factors, including the degree of relation and age of stroke onset in the family member. A family history of stroke at a young age may indicate a stronger genetic predisposition. And finally, the number of affected family members: Having multiple family members with stroke suggests a potentially higher genetic risk. Studies indicate that a family history of stroke can double your risk compared to those without a family history (Mozaffarian et al., 2015; Błaż et al., 2021; Huynh et al., 2023).

Stroke and marital status

Some studies suggest that unmarried individuals (including those who are never married, divorced, or widowed) may have a higher risk of stroke compared to those who are married (Andersen and Olsen, 2018; Liu et al., 2018). This can be attributed to lifestyle factors because unmarried individuals may be more likely to engage in unhealthy behaviors such as smoking, physical inactivity, or poor diet, which contribute to stroke risk. Social support is one of the essential factors because marriage may provide social support and encouragement for healthy habits, which can benefit cardiovascular health. Unmarried individuals may have higher stress levels because spouses can offer emotional support and help manage stress, which can be a risk factor for stroke, and married people might have better access to healthcare and preventive screenings due to spousal encouragement or shared insurance plans (Andersen and Olsen, 2018; Liu et al., 2018).

Stroke and education

Previous studies showed that there is a well-established association between stroke incidence and educational attainment. Prior literature shows that individuals with

lower educational attainment have a higher risk of stroke compared to those with higher education levels (Che et al., 2020; Khan et al., 2021; Wan et al., 2023). The higher incidence may be attributed to socioeconomic factors, which show that individuals with lower education often have lower socioeconomic status, which can limit access to quality healthcare, healthy food options, and safe living environments. These factors contribute to risk factors such as high blood pressure, diabetes, and obesity. Lifestyle choices are also an essential factor because individuals with lower education may be more likely to smoke, have a less healthy diet, and be less physically active, all of which increase stroke risk. Finally, individuals with higher education levels have better knowledge about healthy behaviors and stroke risk factors, which empowers them to make informed choices to reduce their risk.

Stroke and living in an urban or rural area

The stroke incidence might be lower in rural areas than in urban areas. However, the chances of dying from a stroke are higher in rural areas compared to urban areas (Kapral et al., 2019; Khan et al., 2021; Wan et al., 2023). This can be attributed to the fact that individuals in rural areas have limited access to healthcare and lower awareness of stroke risk factors: People living in rural areas may have lower awareness of stroke risk factors and warning signs. This can delay seeking medical attention during a stroke.

Stroke diagnosis

Early diagnosis of stroke is critical to minimize the probability of brain damage because cells begin to die within minutes of a stroke due to lack of oxygen. Therefore, early intervention helps limit this damage by restoring blood flow as quickly as possible, which results in improved recovery, reduced long-term disability, and saving the patient's life (GBD 2019 Stroke Collaborators, 2021). Early and precise stroke localization is vital for providing the best healthcare to patients.

Several imaging techniques are used to evaluate the brain's circulatory system, which is efficient in diagnosing stroke. These imaging modalities include the following:

1. Computed tomography (CT): CT scan is the gold standard initial imaging test for stroke diagnosis, which is often done right after a suspected stroke. A brain CT scan can show whether there is bleeding in the brain or damage to the brain cells from a stroke (Figure 1). CT perfusion has a sensitivity of 80% and specificity for diagnosing ischemic stroke on average (Xin and Han, 2016).
2. Magnetic resonance imaging (MRI): MRI images may identify changes in brain tissue and damage to brain cells. They can also help differentiate between ischemic stroke and other conditions that mimic stroke symptoms. They can also detect damage to the brain tissue even if it is not yet visible on a CT scan (Figure 2). MRI had 89% and 95% sensitivity and specificity, respectively (Krag et al., 2023).
3. Digital subtraction angiography (DSA): DSA is an imaging technique for blood vessels.
4. Nuclear medicine techniques, including positron emission tomography, are used to look for narrowed blood vessels in the neck, an aneurysm, or an AVM (or tangled blood vessels) in the brain.



Figure 1: CT brain showing ischemic stroke. Abbreviation: CT, computed tomography.

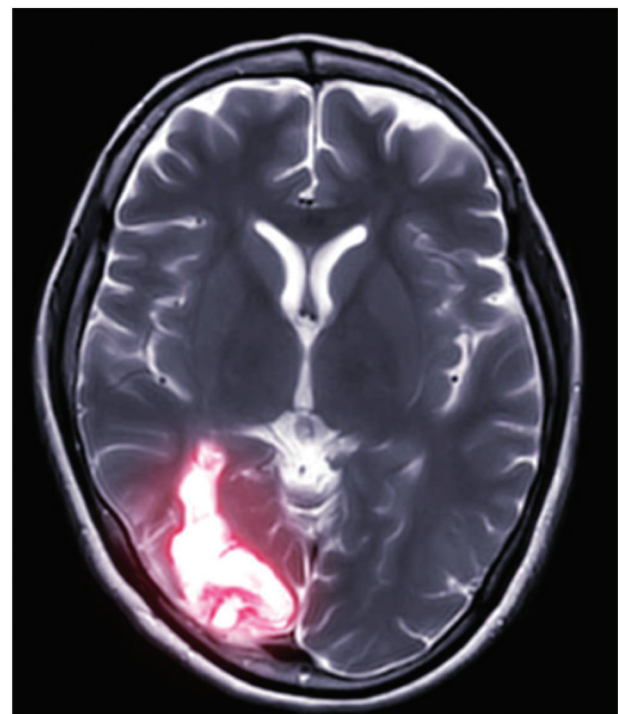


Figure 2: MRI image illustrating stroke and tissue damage. Abbreviation: MRI, magnetic resonance imaging.

The most appropriate diagnostic tests will depend on the specific situation of the patient and the suspected type of stroke. A combination of these techniques may be used to diagnose a stroke and determine the best course of treatment.

Stroke treatment

Stroke treatment focuses on two main goals: minimizing brain damage and preventing further complications. The specific treatment approach depends on the type of stroke (ischemic or hemorrhagic) and the time elapsed since stroke onset. The main treatment options are as follows (Khan et al., 2021; Wan et al., 2023):

- i. Ischemic stroke:
 - Thrombolytic therapy (clot busters): These medications can dissolve blood clots, blocking blood flow to the brain if administered within a narrow time window (typically 3–4.5 h) after stroke onset. They are highly effective in improving outcomes and reducing disability.
 - Endovascular thrombectomy: This minimally invasive procedure involves threading a catheter through an artery to reach the clot and remove it mechanically. It is most effective for large clots in major arteries and can be used within a longer time window (up to 24 h) compared to clot-busting medications.
 - Other treatments: Supportive measures such as blood pressure control, oxygen therapy, medications to prevent blood clots, and fluid management are crucial for stabilizing the patient and promoting recovery.
- ii. Hemorrhagic stroke:
 - Blood pressure control: Medications are used to lower blood pressure carefully to prevent further bleeding in the brain.
 - Surgical intervention: In some cases, surgery may be necessary to remove the blood clot or repair a ruptured blood vessel.
 - Supportive measures: Similar to ischemic stroke, supportive care plays a vital role in ensuring the patient's stability and promoting recovery.
- iii. Stroke rehabilitation:

Following the initial treatment phase, stroke rehabilitation is essential for regaining lost function and improving a patient's quality of life. Rehabilitation may involve (Bindawas and Vennu, 2016; Khan et al., 2021; Wan et al., 2023):

 - Physical therapy: To improve mobility, coordination, and balance.
 - Occupational therapy: To relearn daily living skills such as dressing, bathing, and eating.
 - Speech therapy: To address communication difficulties caused by stroke.
 - Cognitive rehabilitation: To improve memory, thinking, and problem-solving skills.
- iv. Rehabilitation process:
 - Early intervention: Starting rehabilitation as soon as possible after a stroke is crucial. The brain has a remarkable capacity for plasticity, meaning it can reorganize and form new connections even after

damage. Early intervention capitalizes on this neuroplasticity to maximize recovery potential.

- Individualized approach: No single rehabilitation program fits all. Rehabilitation plans are tailored to address the specific needs and goals of each stroke survivor.
- Multidisciplinary team: A rehabilitation team typically consists of physical therapists, occupational therapists, speech therapists, psychologists, and social workers. Each team member plays a crucial role in supporting the patient's recovery journey.
- Long-term commitment: Stroke rehabilitation can be a long-term process, often lasting months or even years. Commitment and perseverance are essential to achieve optimal outcomes.

In general, stroke and rehabilitation are intertwined. While stroke can be a life-altering event, effective rehabilitation can empower survivors to regain function, improve their quality of life, and live as independently as possible.

Stroke and disability

Stroke may lead to loss of function because it can damage brain tissues responsible for various functions such as movement, speech, cognition, and sensation. This damage can lead to disabilities that affect a person's ability to perform daily activities independently (Khan et al., 2021; Wan et al., 2023). Therefore, stroke-related disabilities can be physical, cognitive, or communicative.

- a. Physical disabilities: Paralysis, weakness, balance problems, and impaired coordination can make it difficult to walk, dress, or perform self-care tasks.
- b. Cognitive disabilities: Memory loss, difficulty concentrating, and problems with reasoning and judgment can affect a person's ability to manage finances, make decisions, or follow instructions.
- c. Communicative disabilities: Speech difficulties, swallowing problems, and aphasia (language impairment) can hinder communication and social interaction.

Stroke prevention

Preventive medicine plays a vital role in reducing the risk of stroke. Proper identification and management of risk factors are crucial in successful stroke reduction efforts. These factors are explained in Table 1.

Table 1: Stroke risk factors and methods of prevention (GBD 2019 Stroke Collaborators, 2021; Rahman et al., 2021).

Risk factors	Methods
Hypertension	Control blood pressure
Cholesterol	Control cholesterol levels
Diabetes	Effective diabetes management
Atrial fibrillation	Heartbeat control
Smoking	Smoking cessation
Obesity	Maintaining a healthy weight

Challenges and opportunities

The high prevalence of stroke-induced disability presents significant challenges for healthcare systems globally and in Saudi Arabia. Rehabilitation services are crucial for helping stroke survivors regain independence and improve their quality of life. However, access to these services can be limited, particularly in resource-constrained settings. Additionally, raising public awareness about stroke prevention and risk factors is essential for reducing the overall burden. Combating stroke-related disability requires a multi-pronged approach. Primary prevention strategies that focus on controlling risk factors such as hypertension and diabetes are crucial. Furthermore, investing in stroke prevention campaigns and promoting healthy lifestyles can significantly reduce stroke incidence. Additionally, strengthening healthcare systems to provide accessible and comprehensive rehabilitation services for stroke survivors is essential. This includes physical, occupational, and speech therapy to help individuals regain lost skills and reintegrate into society.

CONCLUSION

Stroke is a significant public health concern with devastating consequences. Its link to disability and morbidity necessitates preventive measures, early intervention, and effective rehabilitation to minimize its impact on individuals and society. Women, especially after menopause, should be aware of their increased stroke risk and take steps to manage their risk factors such as high blood pressure, diabetes, and high cholesterol. Managing established risk factors such as high blood pressure, diabetes, and cholesterol remains the

cornerstone of stroke prevention for all individuals, regardless of profession. Workplace wellness programs that promote healthy lifestyles and stress management can benefit employees in any occupation.

Stroke is a devastating event that can impact individuals, families, and healthcare systems. By acknowledging the global and Saudi Arabian burden of stroke-related disability, we can prioritize preventative measures and invest in comprehensive rehabilitation programs. This collaborative effort is critical for ensuring stroke survivors can regain independence and live fulfilling lives. To achieve this goal, the following are essential: (i) standardized data collection: a nationwide stroke registry is needed to provide more accurate and consistent data on stroke incidence and disability in Saudi Arabia and (ii) long-term outcomes: studies are required to understand the long-term impact of stroke on survivors' lives, including their disability levels, healthcare needs, and quality of life. Overall, stroke presents a growing public health challenge in Saudi Arabia. While the incidence may be lower than in some high-income countries, it is rising. More research is needed to gain a clearer picture of stroke-related disability and develop effective prevention and treatment strategies. More research is required to solidify the connection between specific occupations and stroke incidence. This research should consider factors such as job demands, work environment, and individual health behaviors.

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